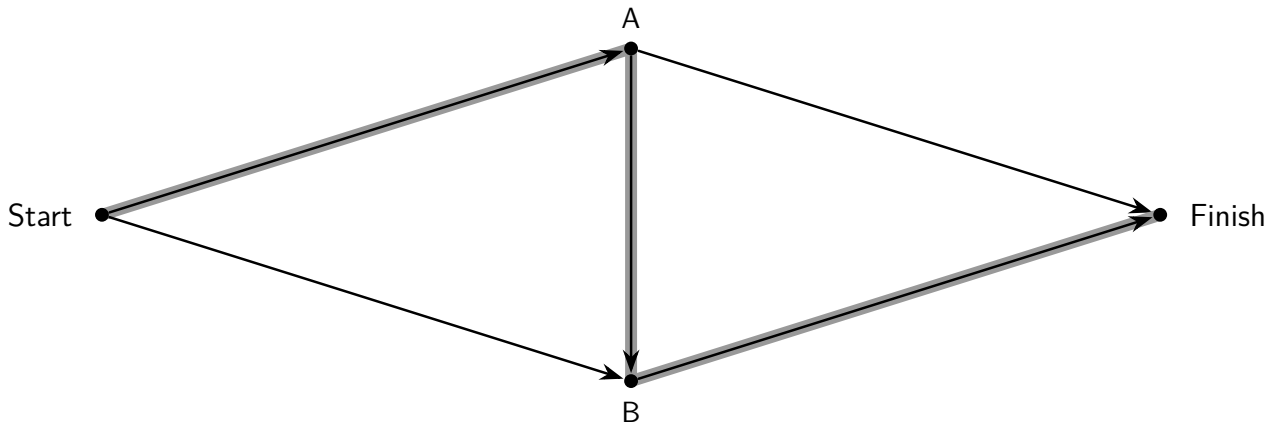
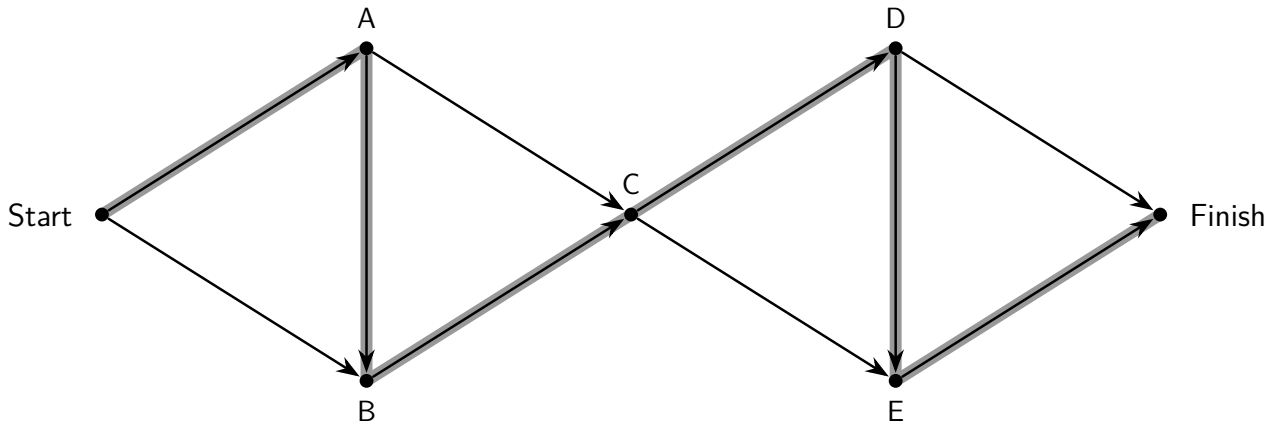
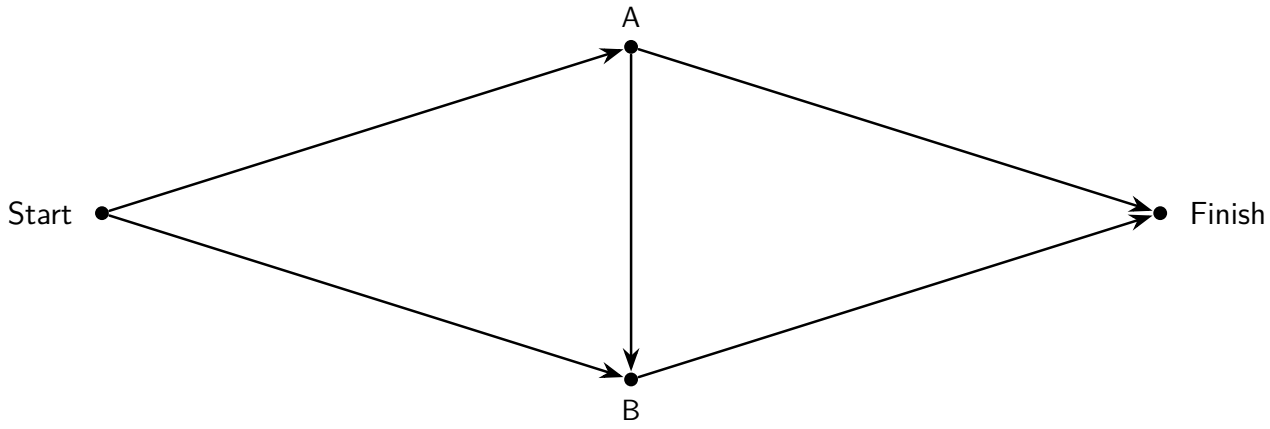
A route follows arrows from Start to Finish without visiting a dot twice. Routes may meet at dots, but each arrow can belong to only one route in a collection. Use a different color or line pattern for each route. A marker closes one arrow.

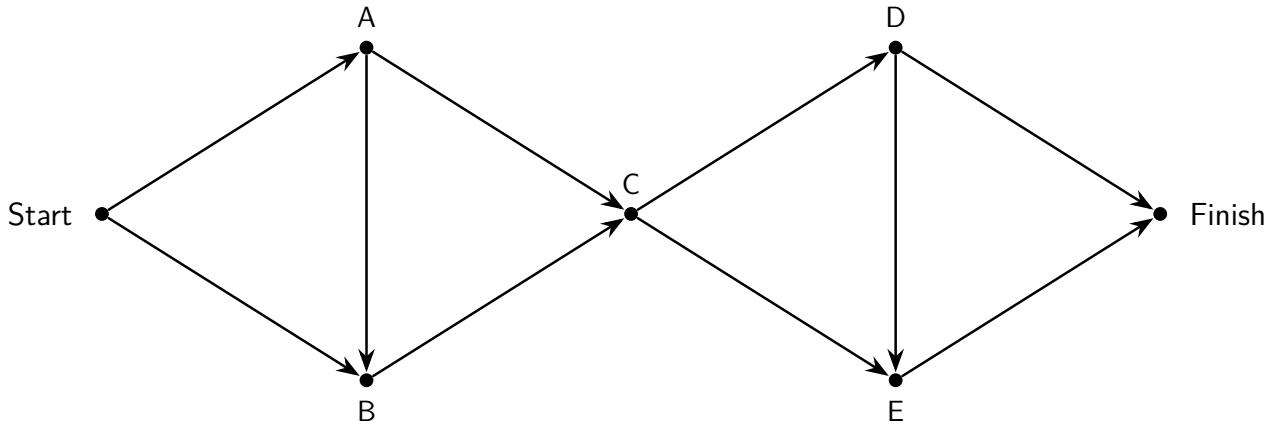
Problem 1: The thick arrows form reserved routes. No more routes fit beside them. Replace them with a largest collection on each board.



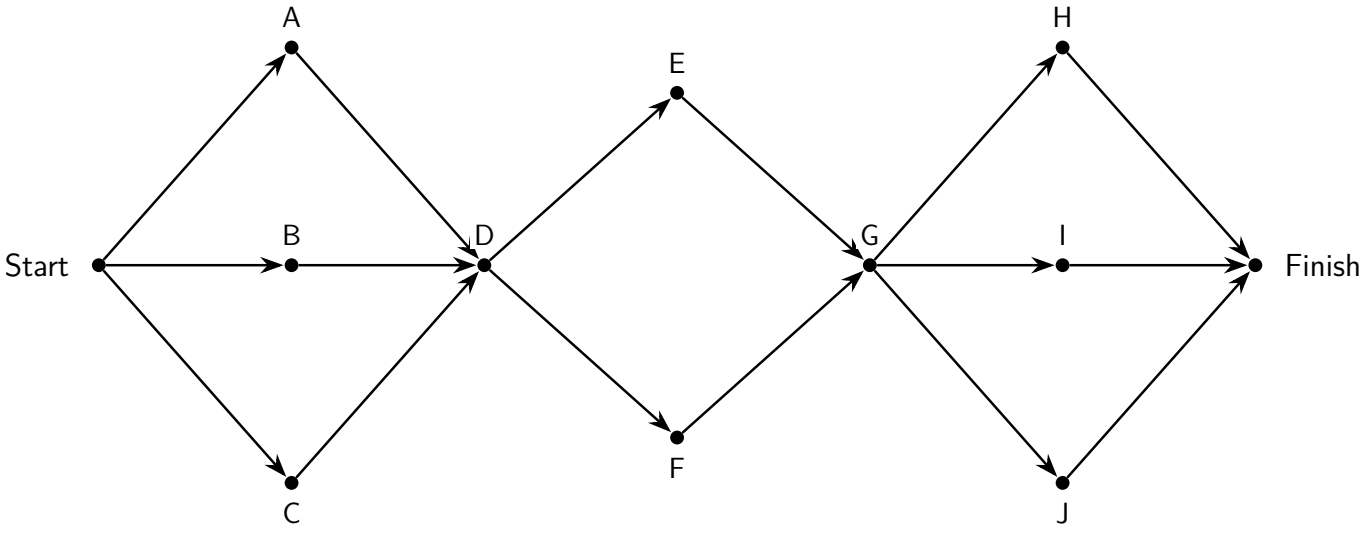


Extra workspace for Problem 1.

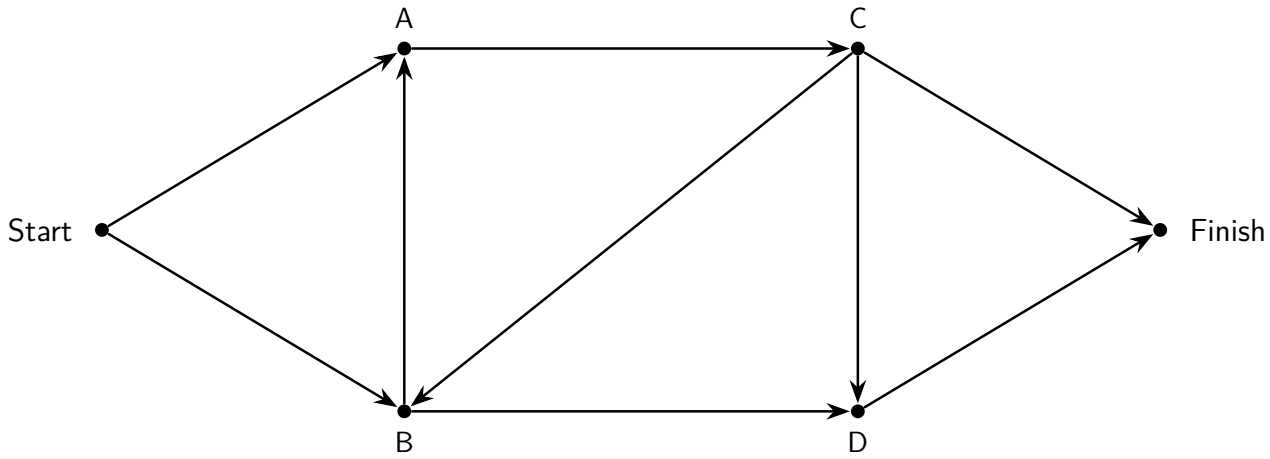


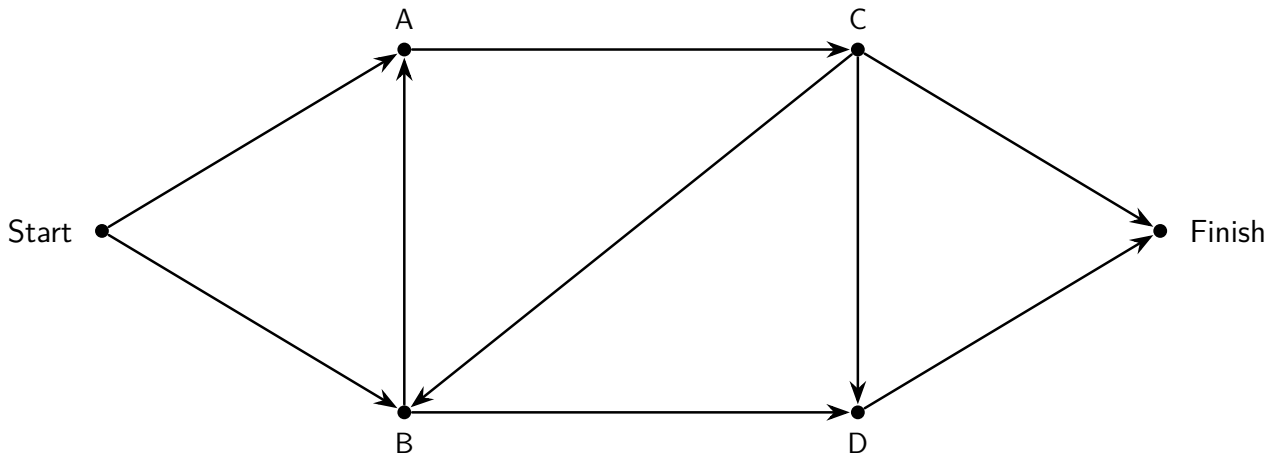


Problem 2: Find a largest route collection and a smallest set of closing arrows. Explain why your closing arrows stop every route, and how you know your collection cannot be larger.

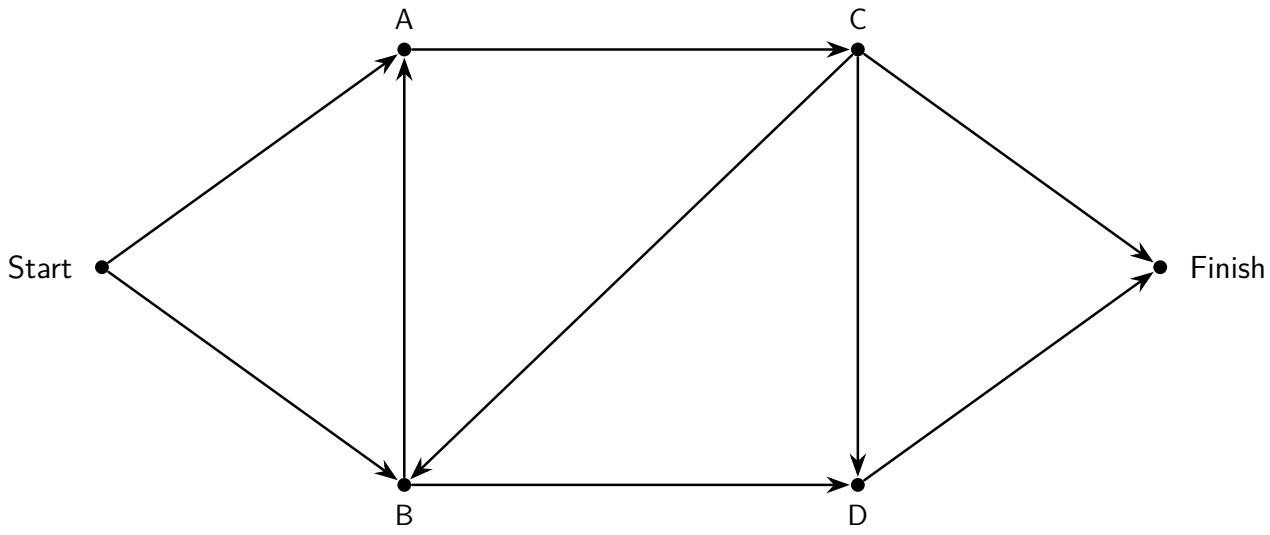


Problem 3: Split the dots into two sides, with Start on one side and Finish on the other. The arrows pointing from the Start side to the Finish side are called the cut arrows. Find two different splits with exactly two cut arrows and at least one arrow pointing back into the Start side. Explain why closing the cut arrows stops all travel.

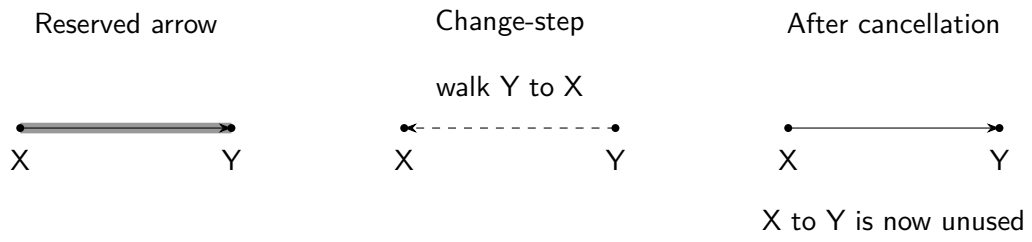




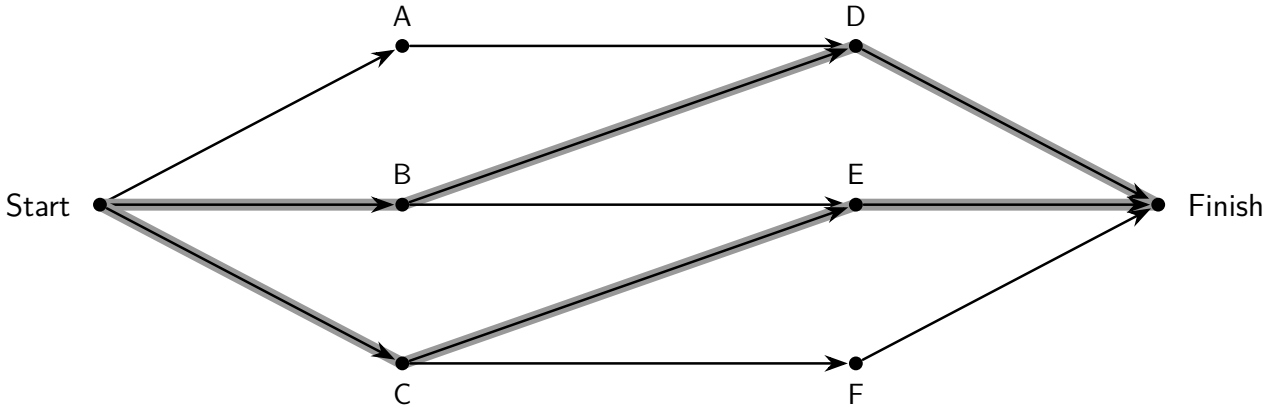
Problem 4: Find a smallest set of closing arrows and a route that uses two of those arrows. Can that route belong to a largest collection on this board? Explain.

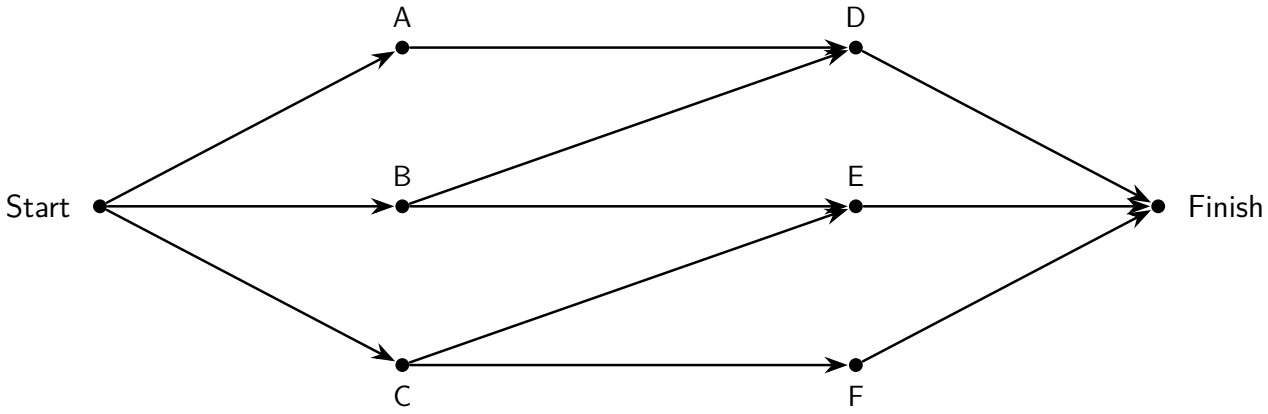


A change-walk follows an unused arrow forward or a reserved arrow backward. A forward step reserves its arrow; a backward step cancels its reservation. A change-walk visits each dot at most once. Final routes follow the printed arrows.

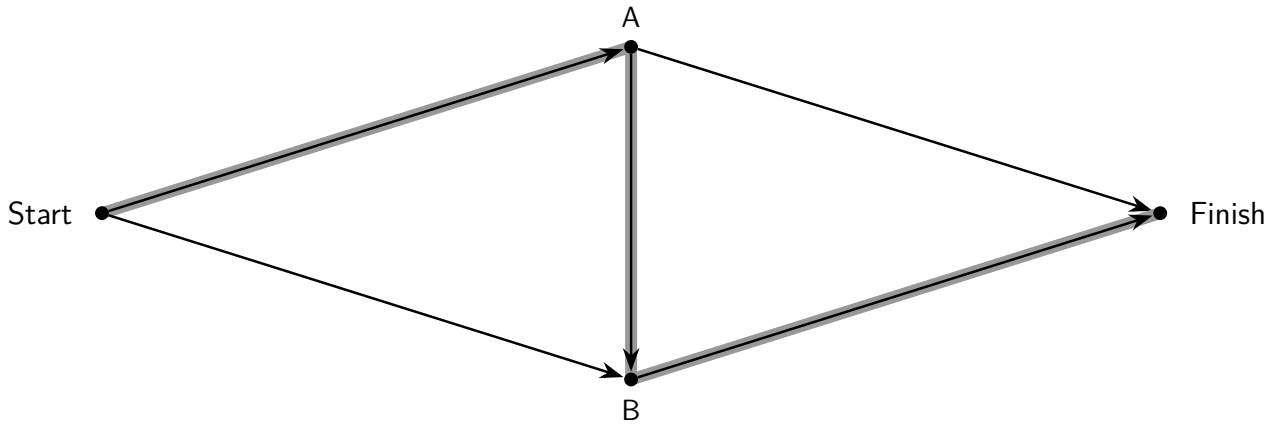


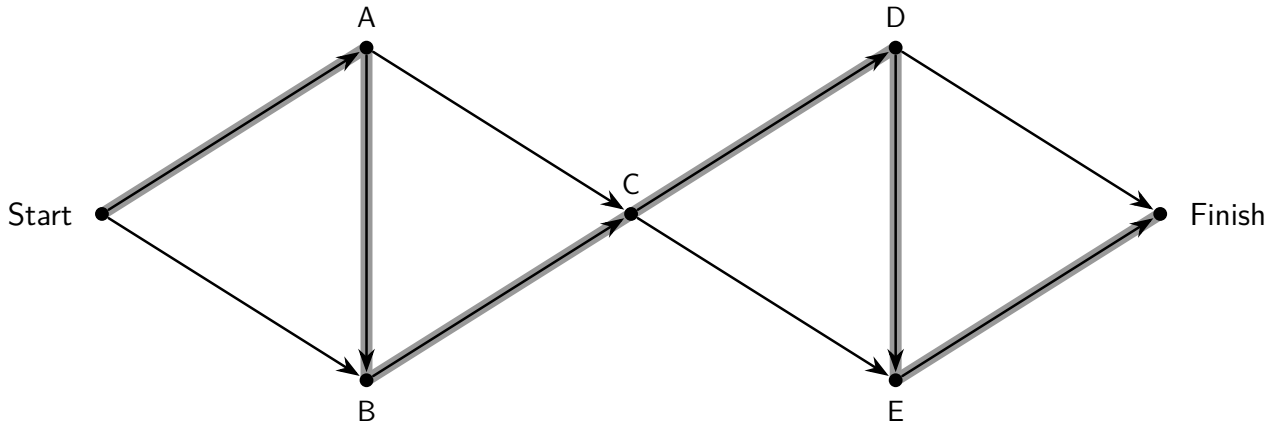
Problem 5: Find a change-walk from Start to Finish that increases the number of routes by one, on this board and on both boards of Problem 1. Draw each resulting route collection.



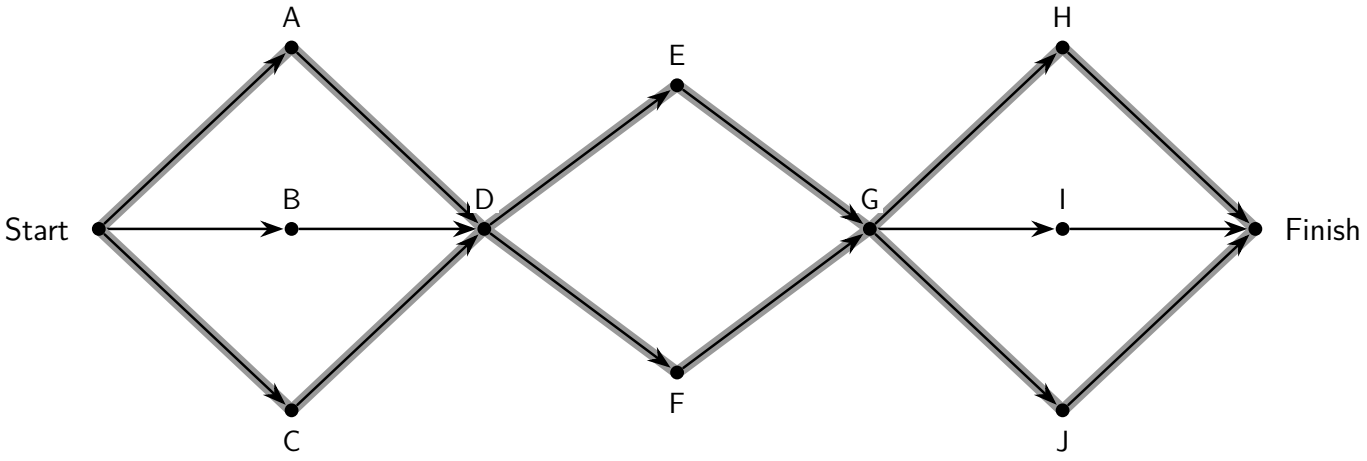


Extra workspace for Problem 5.





Problem 6: The thick arrows form two reserved routes. Find every dot you can reach from Start by change-walk steps. Find a smallest set of closing arrows, and explain why it matches the route collection.



Problem 7: On any finite board with these rules, suppose no change-walk from Start can reach Finish. Must the reserved route collection already be largest? Explain why, or draw a board where it is not.